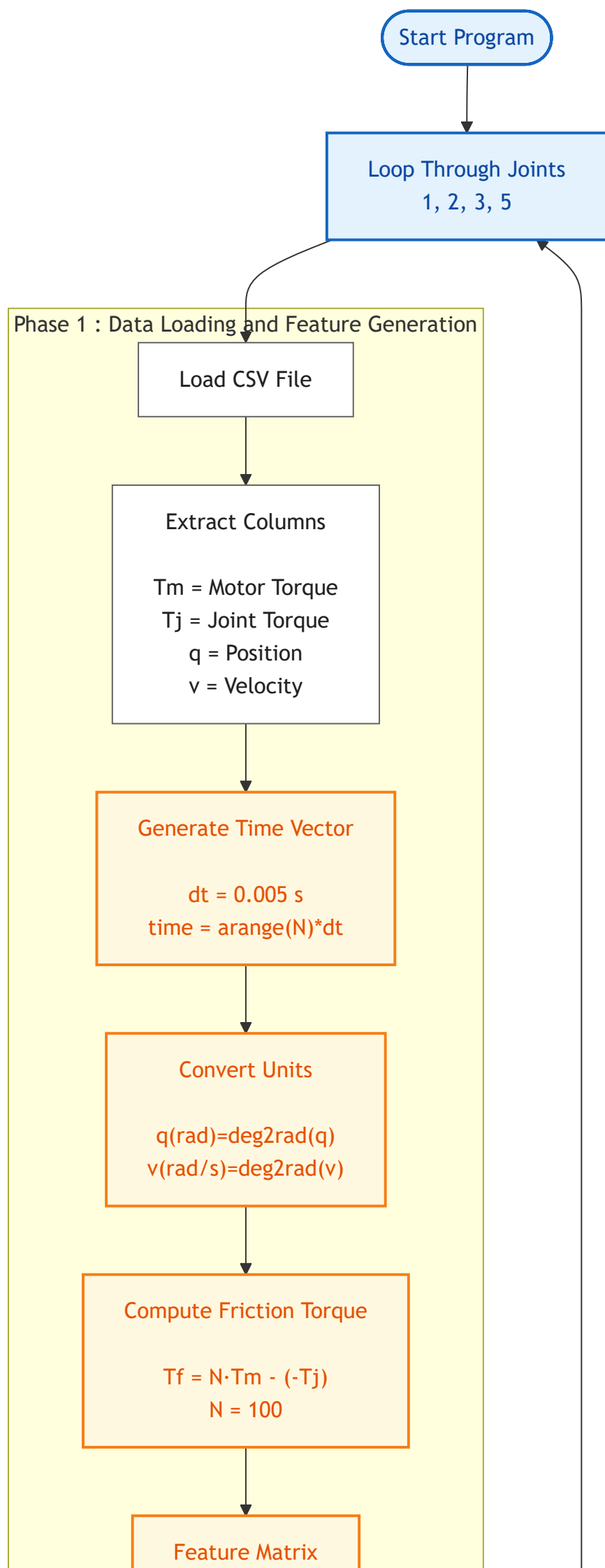




$X = [q, v]$

Target = T_f

Phase 2 : Raw Data Visualization

Plot Raw Time Series

Motor Torque vs Time

Joint Torque vs Time

Velocity vs Time

Position vs Time

Phase 3 : Random Dataset Split

Random Split

20% Test Set

Remaining 80%

Split Remaining Data

70% Train

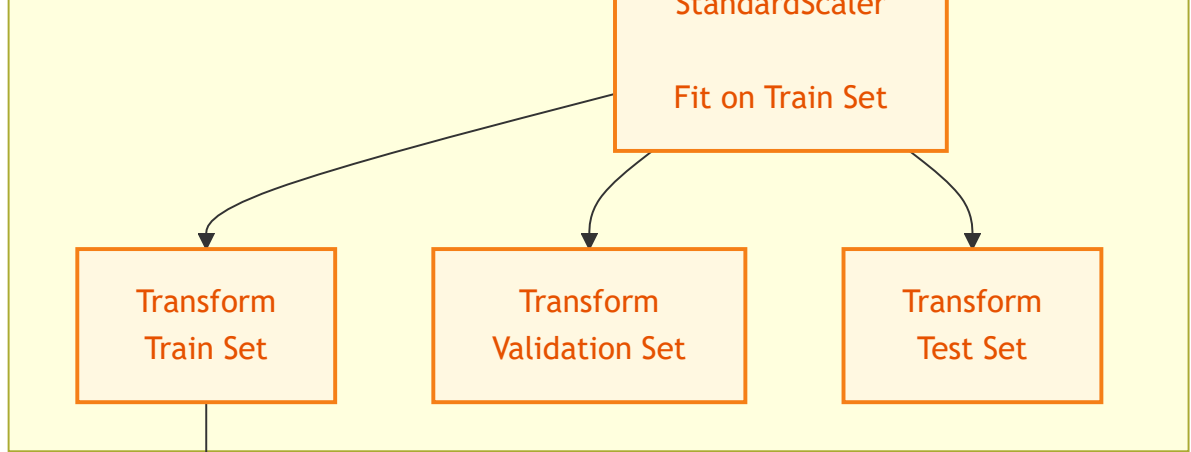
10% Validation

Sort Test Set by Time

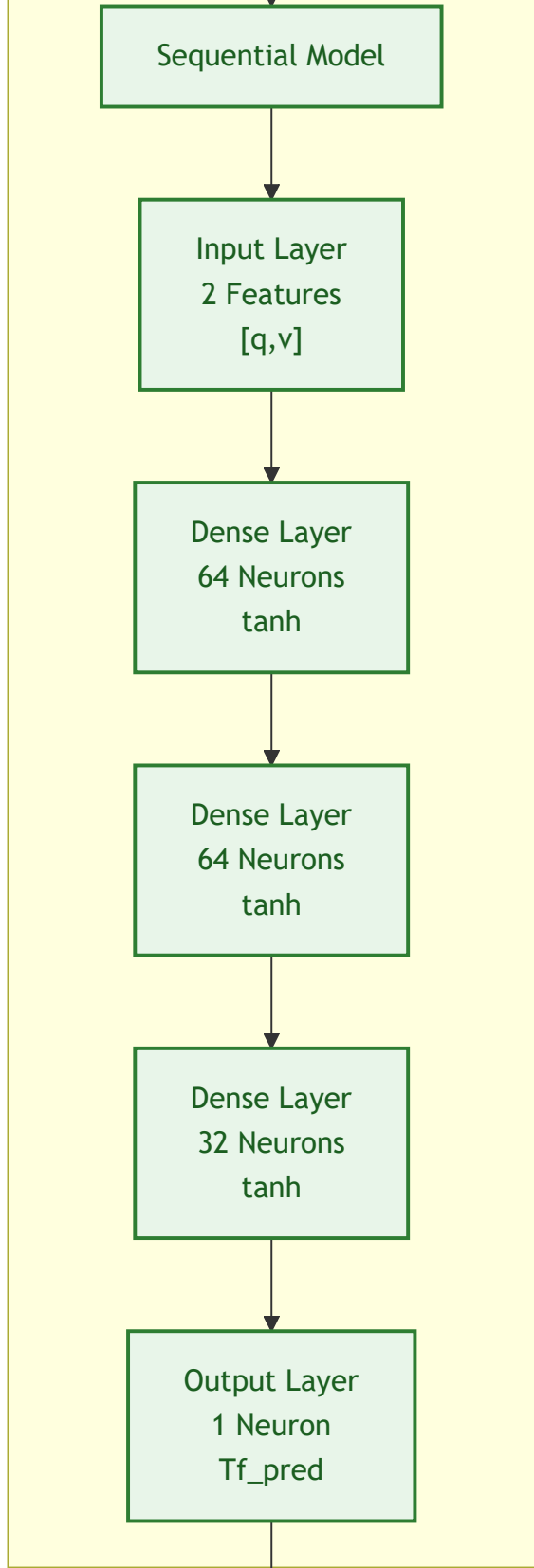
For Smooth Visualization

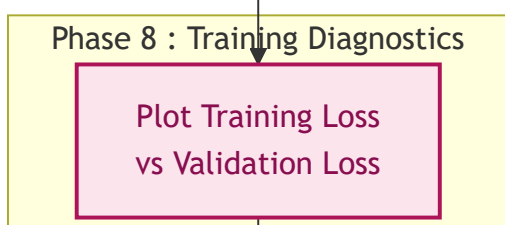
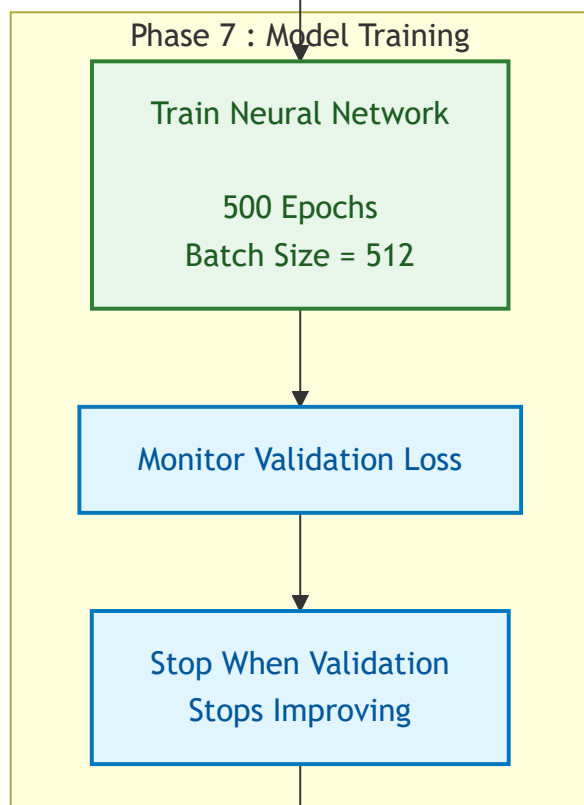
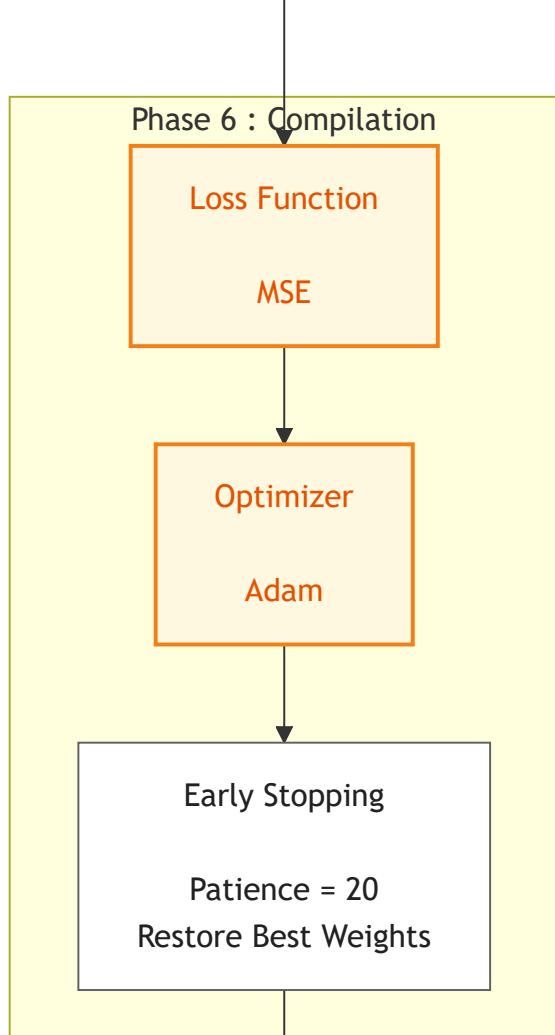
Phase 4 : Feature Scaling

StandardScaler



Phase 5 : Neural Network Architecture





Phase 9 : Prediction

Predict Friction Torque
on Test Set

Phase 10 : Performance Evaluation

Compute RMSE

Compute MAE

Compute R² Score

Print Model Metrics

Phase 11 : Test Set Diagnostics

Actual vs Predicted
Time Domain Plot

Actual vs Predicted
Scatter Plot

Residual Calculation

Residual = Actual -
Predicted

Residual vs Time

Residual Histogram

